1. ***This is the start of a multipart question. The information below pertains to question 1 to question 8.***

   The data set `wage2` contains data on working men in the US from the 1980 National Longitudinal Survey. You can load the data into R using the `wooldridge` library using:

   ```
   library(wooldridge)
   dat = wooldridge::wage2
   ```

   The main outcome of interest is the variable *wage*, which is monthly earnings in dollars. What is the median of the variable *wage* in the dataset?

   a. 957.95
   b. 905.00
   c. 925.00
   d. 1000.00
   e. None of the above

2. How many missing values or NA's are there for the variable *meduc* in the dataset?

   a. 83
   b. 194
   c. 0
   d. 78
   e. None of the above

3. Consider the following linear regression model.

$$\log(wage) = \beta_0 + \beta_1(educ) + \beta_2(exper) + \beta_3(exper^2) + \beta_4(tenure) + \beta_5(married) + \beta_6(black) + \beta_7(south) + \beta_8(urban) + \beta_9(meduc) + \varepsilon$$

   The model includes controls for years of education, years of work experience and experience squared, and years with the current employer (*tenure*). The model also controls for mother's education (*meduc*). Finally, (*married, black, south, urban*) are indicator variables.
   Estimate the model above with OLS. What is the estimated coefficient on *educ*?

   a. 0.06541
   b. 0.05982
   c. 0.06313
   d. 0.07012
   e. 0.05433

4. What is the residual standard error ($\hat{\sigma}$) of the model?

   a. 0.1312
   b. 0.3623
   c. 0.3601

d. 111.152
e. None of the above

5. Suppose we wanted to create a confidence interval for one of the parameters of the model. If we wanted to create a 98% confidence interval, what critical value would we use?

   a. 2.3263
   b. 2.0569
   c. 2.5816
   d. 2.3304
   e. None of the above

6. Consider the null hypothesis that $H_0 : \beta_{south} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

   a. Reject the null at the 0.1% level
   b. Reject the null at the 1% level
   c. Reject the null at the 5% level
   d. Reject the null at the 10% level
   e. Fail to reject the null at the 10% level

7. Consider the null hypothesis that $H_0 : \beta_{educ} = 0.05$. What is the p-value associated with this hypothesis test?

   a. 0.0266
   b. 0.0531
   c. 0.9469
   d. 0.0000
   e. None of the above

8. What is the covariance between the estimate of the parameter on *exper* and the parameter on $exper^2$?

   a. -7.4233×10^{-6}
   b. 7.4233×10^{-6}
   c. -5.8152×10^{-4}
   d. -3.2770×10^{-4}
   e. None of the above

9. ***This is the start of a multipart question. The information below pertains to question 9 to question 15.***

   The data set `crime1` contains data on young men in California who were arrested at least once before 1986. You can load the data into R using the `wooldridge` library using:

```
library(wooldridge)
dat = wooldridge::crime1
```

We would like to study what are the main determinants of being arrested in 1986 using a linear probability model. To answer this question you will need to add a new variable to the data set called *arrested* which should equal 1 if the man was arrested at least once in 1986 (i.e., *narr86* > 0) and zero otherwise.
What percent of the population were *arrested*?

a. 72.29%
b. 40.48%
c. 27.71%
d. 7.21%
e. None of the above

10. Estimate a linear probability model on *arrested*.
Include in your model controls for the proportion of prior arrests that led to a conviction (*pcnv*), months spent in prison in 1986 (*ptime86*), the number of quarters employed in 1986 (*qemp86*), and *black*. Your model should have 5 parameters including the intercept.
What is the adjusted R-squared from the estimated model?

a. 0.06053
b. 0.05915
c. 0.04412
d. 0.07230
e. 0.05121

11. What is the correct interpretation of the parameter on *qemp86*? Holding fixed the variables in the model, one additional quarter of employment changes the probability of being arrested by . . .

a. –3.8155 percent
b. –0.0382 percentage points
c. –3.8155 percentage points
d. –38.155 percentage points
e. None of the above

12. What is the estimate of the lower bound of the 95% confidence interval for the parameter on *qemp86*?

a. –0.04707
b. –0.02754
c. –0.04914
d. –0.04878
e. None of the above

13. Consider the null hypothesis that $H_0 : \beta_{pcnv} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

a. Reject the null at the 0.1% level
b. Reject the null at the 1% level
c. Reject the null at the 5% level
d. Reject the null at the 10% level
e. Fail to reject the null at the 10% level

14. Re-estimate the model adding legal income in 1986 (*inc86*) as an additional control. What is the percentage change in the parameter on *qemp86* when we control for income?

a. –92.23%
b. 92.23%
c. –12.87%
d. 7.77%
e. None of the above

15. Re-estimate the model adding legal income in 1986 (*inc86*) as an additional control. What is the variance inflation factor for the parameter on *qemp86* when we control for income? (note: if you are calculating the variance inflation factor from the formula you need to ensure that the same observations are used in the different regressions)

a. 2.0362
b. 1.0833
c. 2.1398
d. 0.4673
e. None of the above

Solution Key

Q	Ans	Value	How / trap
1	**b**	905	`median(dat$wage)`; 957.95 is the mean
2	**d**	78	`sum(is.na(dat$meduc))`; 83 = *brthord*, 194 = *feduc*
3	**c**	0.06313	`log(wage)` or `lwage`; lm drops 78 rows, $n = 857$
4	**b**	0.3623	`summary(m)$sigma`; 0.1312 = $\hat{\sigma}^2$, 111.152 = SSR (`deviance(m)`)
5	**e**	2.3308	`qt(0.99, df.residual(m))`, df = 857−10 = 847. 2.3304 uses 935 − 10; 2.3263 is `qnorm`
6	**b**	$p = 0.0048$	$0.001 < p < 0.01$
7	**b**	0.0531	`linearHypothesis(m, "educ = 0.05"), Pr(>F)`; 0.0266 is one-sided
8	**a**	-7.42×10^{-6}	`vcov(m)["exper","I(exper^2)"]`
9	**c**	27.71%	`arrested = as.numeric(narr86 > 0)`
10	**b**	0.05915	`summary(m1)$adj.r.squared`; 0.06053 is R^2
11	**c**	−0.0382	LPM: $\times 100$ = percentage points
12	**d**	−0.04878	`confint(m1, "qemp86")`; −0.04707 is the 90% bound
13	**a**	$p \approx 10^{-13}$	$p < 0.001$
14	**a**	−92.23%	`(new - old) / old * 100`; 7.77% is just the ratio
15	**c**	2.1398	`car::vif(m2)`; 2.0362 is the VIF of *inc86*

R code

```
# ECON 5021 Unit 2 Data Analysis Practice B: 26101200003 - solution key
# Key: 1b 2d 3c 4b 5e 6b 7b 8a 9c 10b 11c 12d 13a 14a 15c

library(wooldridge)
library(car)

# ---- Q1-8: wage2 ----
dat = wooldridge::wage2

# Q1. What is the median of the variable wage in the dataset?
median(dat$wage)                                  # 905     (b)

# Q2. How many missing values or NA's are there for the variable meduc?
sum(is.na(dat$meduc))                             # 78      (d)

# Q3. Estimate the model with OLS. Coefficient on educ?
m = lm(log(wage) ~ educ + exper + I(exper^2) + tenure + married + black + south + urban + meduc,
     data = dat)
summary(m)                                        # 0.06313 (c)   n = 857

# Q4. What is the residual standard error (sigma-hat) of the model?
```

```
summary(m)$sigma                                  # 0.3623  (b)

# Q5. Critical value for a 98% confidence interval?
qt(0.99, df.residual(m))                          # 2.3308  (e)   df = 847; 2.3304 uses the wrong
     df (935 - 10)

# Q6. H0: b_south = 0. Strongest conclusion?
summary(m)$coefficients["south", ]                # p = 0.0048 -> reject at 1% (b)

# Q7. H0: b_educ = 0.05. p-value?
linearHypothesis(m, "educ = 0.05")                # Pr(>F) 0.0531  (b)   F = t^2, same p as the t
    -test

# Q8. Covariance between the estimates on exper and exper^2?
vcov(m)["exper", "I(exper^2)"]                    # -7.4233e-06 (a)

# ---- Q9-15: crime1 ----
dat = wooldridge::crime1
dat$arrested = as.numeric(dat$narr86 > 0)

# Q9. What percent of the population were arrested in 1986?
100 * mean(dat$arrested)                          # 27.71%  (c)

# Q10. LPM of arrested on pcnv, ptime86, qemp86, black. Adjusted R-squared?
m1 = lm(arrested ~ pcnv + ptime86 + qemp86 + black, data = dat)
summary(m1)$adj.r.squared                         # 0.05915 (b)

# Q11. Interpretation of the parameter on qemp86?
coef(m1)["qemp86"]                                # -0.038155 -> -3.8155 percentage points (c)

# Q12. Lower bound of the 95% CI for the parameter on qemp86?
confint(m1, "qemp86", level = 0.95)               # -0.04878 (d)

# Q13. H0: b_pcnv = 0. Strongest conclusion?
summary(m1)$coefficients["pcnv", ]                # p ~ 6e-13 -> reject at 0.1% (a)

# Q14. Add inc86. Percentage change in the parameter on qemp86?
m2 = lm(arrested ~ pcnv + ptime86 + qemp86 + black + inc86, data = dat)
old = coef(m1)["qemp86"]
new = coef(m2)["qemp86"]
(new - old) / old * 100                           # -92.23% (a)

# Q15. VIF for qemp86 when we control for inc86?
vif(m2)["qemp86"]                                 # 2.1398  (c)
```